

RES.6-0.12 Introduction to Probability - Lecture 7-10
 Textbook Problem Sets - Continuous RVs - Section 3.1-35

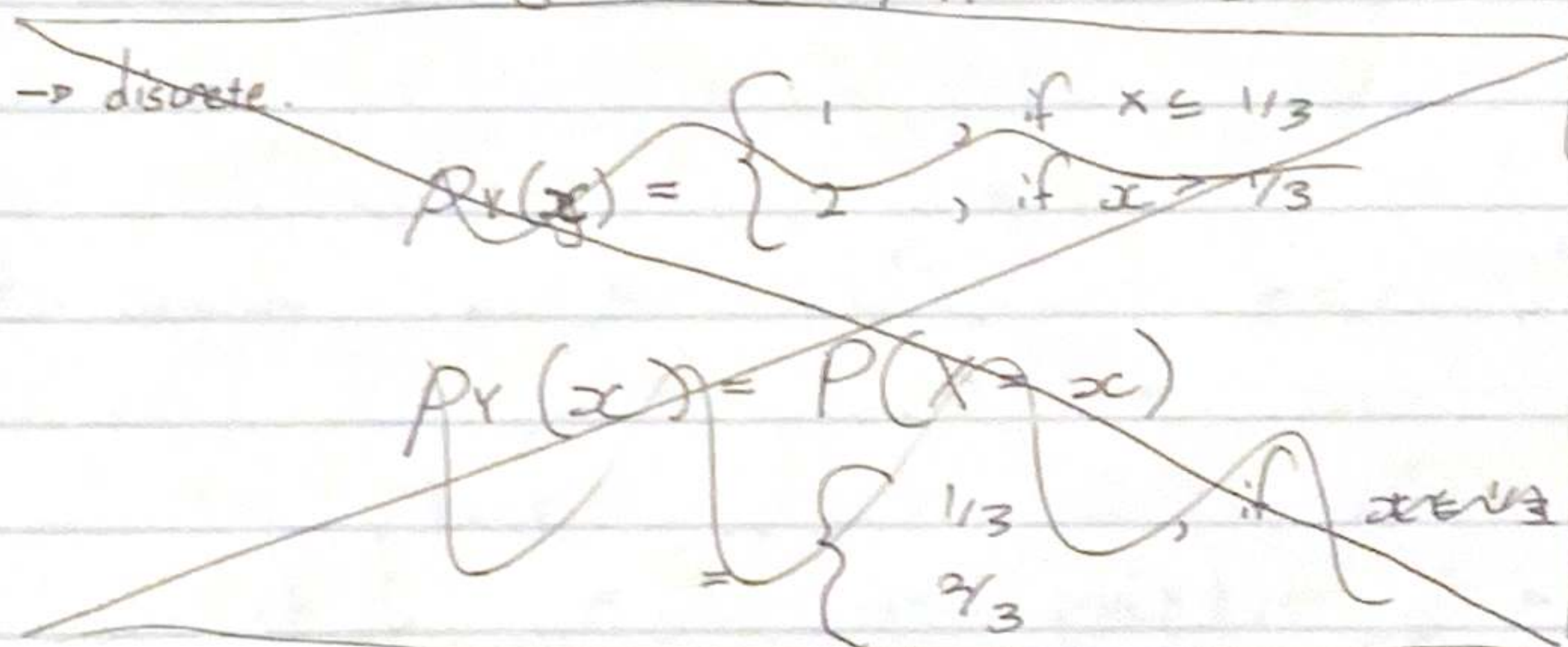
Problem Set Section 3.1

1) $X \sim \text{uniform}[0, 1]$ RV: $Y = g(X)$

$$g(x) = \begin{cases} 1, & \text{if } x \leq 1/3 \\ 2, & \text{if } x > 1/3 \end{cases}$$

$$p_Y(1) = P(X \leq 1/3) = 1/3$$

$$\therefore p_Y(2) = P(X > 1/3) = 1 - p_Y(1) = 2/3$$



$$p_Y(y) = \begin{cases} 1, & \text{if } x \leq 1/3 \\ 2, & \text{if } x > 1/3 \end{cases}$$

$$p_Y(x) = P(X = x)$$

$$= \begin{cases} 1/3, & \text{if } x = 1 \\ 2/3, & \text{if } x = 2 \end{cases}$$

$$E[Y] = \sum_j y \cdot p_Y(y)$$

$$= (1/3)(1)$$

$$= (1)(1/3) + (2)(2/3)$$

$$= 5/3$$

Expected Value Rule: $E[g(x)] = \sum_x g(x) \cdot p_X(x)$
 distributed uniform $[0, 1]$

$$E[g(x)] = \int_0^1 g(x) dx$$

$$\begin{aligned} E[Y] &= \int_0^1 g(x) \cdot f_X(x) dx = \int_0^{1/3} 1 \cdot dx + \int_{1/3}^1 2 \cdot dx \\ &= x \Big|_0^{1/3} + 2x \Big|_{1/3}^1 \\ &= (1/3) - (0) + 2(1) - 2(1/3) \\ &= 5/3 \end{aligned}$$

$$\Rightarrow f_X(x) = \frac{1}{b-a} = \frac{1}{1-0} = 1$$

Each distribution has own way
 to calculate $f_X(x)$ → Exponential: $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

3) Show Expected Value of RV satisfies:

$$E[X] = \int_0^{\infty} P(X > x) dx - \int_0^{\infty} P(X < -x) dx.$$

① $\rightarrow P(X > x) = \int_x^{\infty} f_X(y) dy.$

$y \rightarrow$ dummy to distinguish from x .

to prove: $E[X] = \int_0^{\infty} P(X > x) dx$

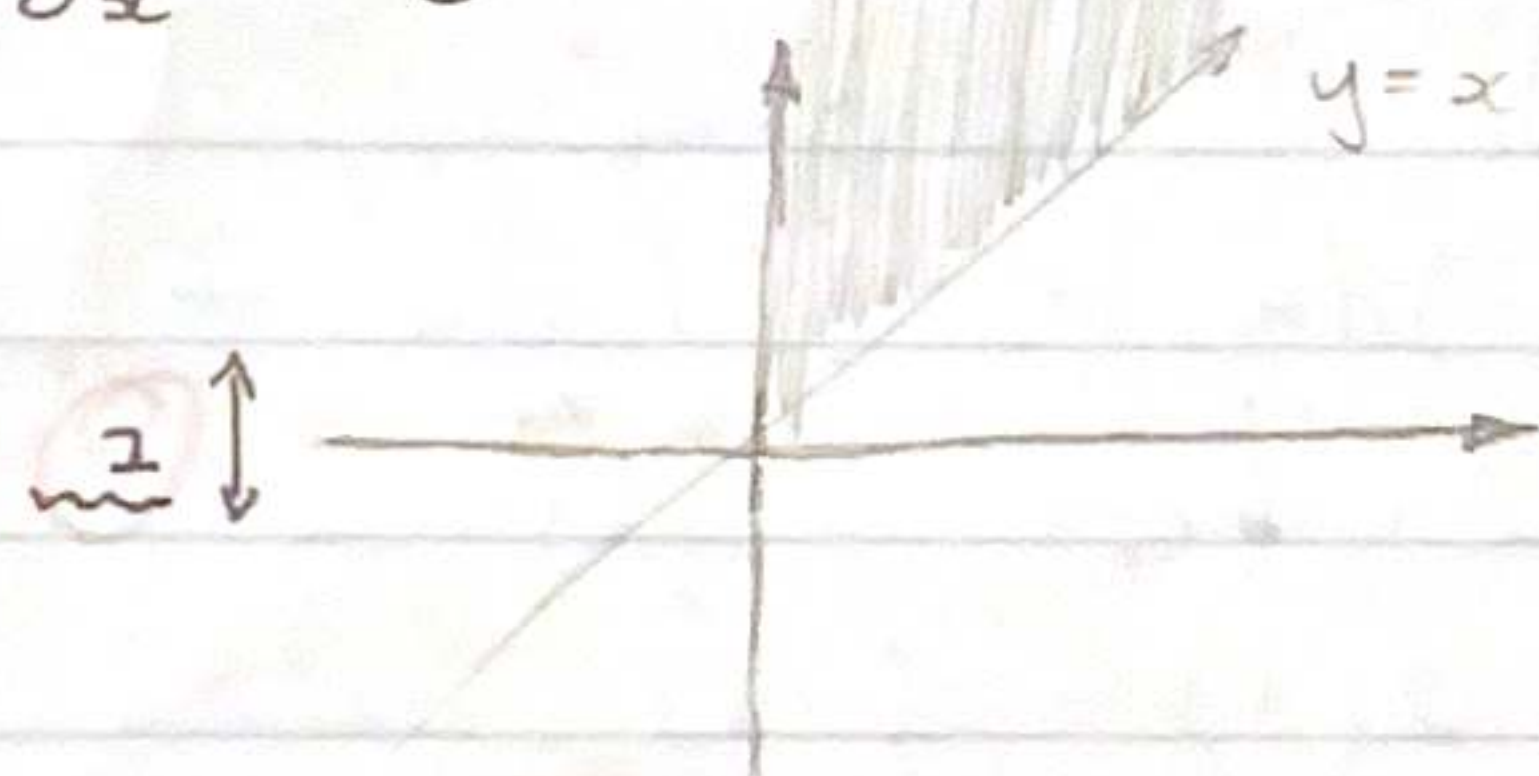
$\hookrightarrow \therefore \int_0^{\infty}$

$\rightarrow E[X] = \int_0^{\infty} P(X > x) dx - \int_0^{\infty} P(X < -x) dx$ assume $x > 0$
 $= \dots - 0$

① $P(X > x) = \int_x^{\infty} f_X(y) dy \Rightarrow E[X] = \int_0^{\infty} \left[\int_x^{\infty} f_X(y) dy \right] dx.$

$\hookrightarrow$ Note: integrate over x & y .

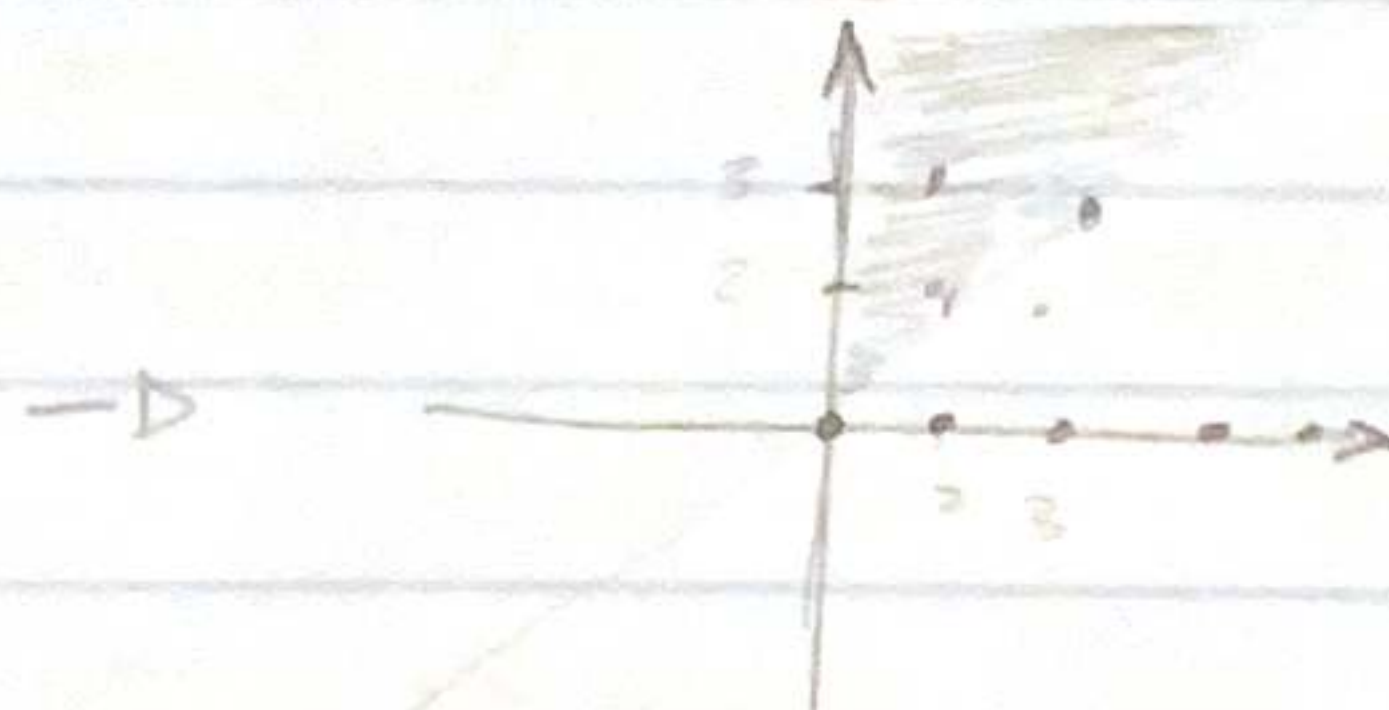
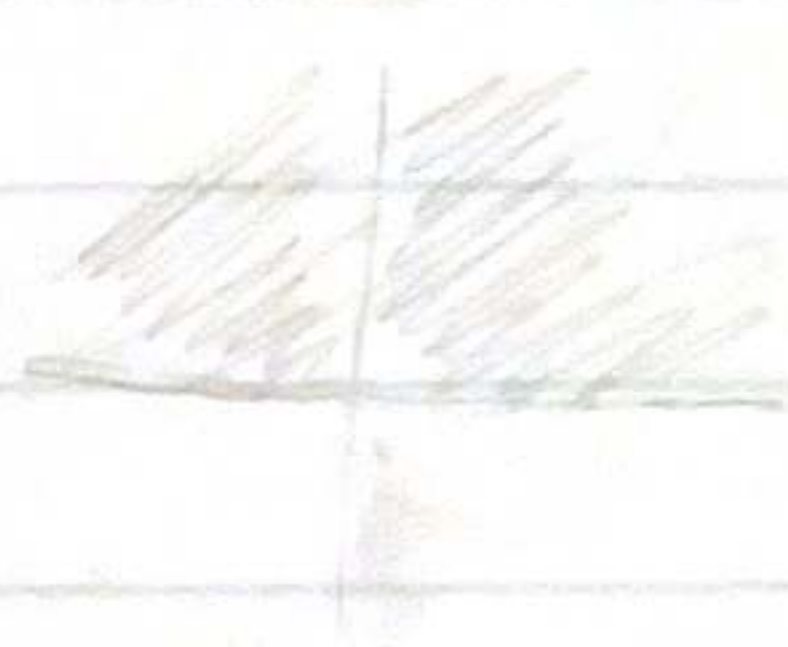
② $\int_0^{\infty} \int_x^{\infty} f_X(y) dy dx \rightarrow x$ runs from 0 to ∞ , and for each x :
 $\rightarrow y$ runs from x to ∞



$0 \leq x < \infty$ and $y \geq x$

③ Switch Order of integration $\rightarrow$ build domain from y then x (instead x then y)

$\rightarrow 0 \leq y < \infty$ and $0 \leq x \leq y$



$(x \leq y)$

$\hookrightarrow \int_0^{\infty} \int_0^y f_X(y) dx dy$

3 con.) ④ $\int_0^\infty \int_0^y f_X(y) dx dy$

$\rightarrow \int_0^y f_X(y) dx = f_X(y) \cdot \int_0^y dx$
 $= f_X(y) \cdot y$
 $\rightarrow$ does not depend on x

$\Rightarrow \int_0^\infty f_X(y) \cdot y dy$ for $\boxed{x > 0}$

$\rightarrow$ by same argument $\int_{-\infty}^0 f_X(y) \cdot (-y) dy$ for $\boxed{x < 0}$

$\therefore E[X] = \int_0^\infty y f_X(y) dy - \int_{-\infty}^0 (-y) f_X(y) dy = \int_{-\infty}^\infty y \cdot f_X(y) dy$

• Section 3.2 Problem Set

5)  $\rightarrow$ Uniform distribution.
 $\hookrightarrow$ Weighted prob based on base wider than apex?

X - distance point to triangle base.
 h - height of triangle

$F_X(x) = P(X \leq x)$

~~$\int_{-\infty}^x \frac{1}{h} dx$~~ $= \int_{-\infty}^x \frac{1}{h} dx$

Area = $\frac{bh}{2}$

$F_X(x) = 1 - P(X > x) = 1 - \frac{A_x}{A} = 1 - \frac{\frac{b(h-x)^2}{2}}{\frac{bh}{2}}$
 $= 1 - \left(\frac{h-x}{h}\right)^2$

while $F_X(x) = 0$ for $x < 0$ & $F(x) = 1$ for $x \geq h$

$\downarrow$ work through
 CDF: $F_X(x) = 1 - \left(\frac{h-x}{h}\right)^2$

PDF: $f_X(x) = \frac{dF_X}{dx}(x) = \frac{d}{dx} \left[1 - \left(\frac{h-x}{h}\right)^2 \right] = \frac{(h-x)^2}{h^2}$
 $= -2 \frac{d}{dx} \frac{x^2}{h} = -2 \frac{2x}{h} = -\frac{4x}{h}$
 $= \frac{2}{h} (2x - 2)$

5 con.) $F_X(x) = 1 - \left(\frac{h-x}{h}\right)^2$

PDF $\rightarrow$ derivative of CDF

$\rightarrow$ Work through of derivation.

$$\begin{aligned} f_X(x) &= \frac{d}{dx} \left(1 - \left(\frac{h-x}{h}\right)^2 \right) \\ &= -\frac{1}{h^2} \cdot \frac{d}{dx} (h-x)^2 \\ &= -\frac{1}{h^2} \cdot \frac{d}{dx} u^2 \quad \text{Let } u = h-x \\ &\quad \frac{du}{dx} = -1 \\ &= -\frac{1}{h^2} \cdot 2(h-x) \cdot (-1) = \frac{2(h-x)}{h^2} \\ &= \frac{2(h-x)}{h^2} \end{aligned}$$

7) a) ~~scribbles~~

$$\begin{aligned} P(a \leq X \leq b) &= \int_a^b f_X(x) dx \\ &= \int_0^r \frac{1}{r} dx \quad \begin{matrix} a=0 \\ b=r \end{matrix} \end{aligned}$$

~~scribbles~~ First calculate CDF: $F_X(x) = \int_0^x \frac{1}{r} dx$

$$\begin{aligned} F_X(x) &= P(X \leq x) = \frac{\pi x^2}{\pi r^2} = \left(\frac{x}{r}\right)^2 \\ &= \int_0^x \frac{\pi x^2}{\pi r^2} dx = \int_0^x \left(\frac{x}{r}\right)^2 dx \\ &= \frac{1}{3} \left(\frac{x}{r}\right)^3 \Big|_0^x = \frac{x^3}{3r^3} \end{aligned}$$

$$F_X(x) = \left(\frac{x}{r}\right)^2$$

$$f_X(x) = \frac{dF_X}{dx} = \frac{d}{dx} \left(\frac{x^2}{r^2}\right) = \frac{1}{r^2} \cdot \frac{d}{dx} (x^2)$$

$$f_X(x) = \begin{cases} \frac{2x}{r^2} & \text{if } 0 \leq x \leq r \\ 0 & \text{otherwise} \end{cases}$$

$$E[X] = \int_0^r x \left(\frac{2x}{r^2}\right) dx$$

$$= \frac{1}{r^2} \int_0^r 2x^2 dx = \frac{2}{r^2} \int_0^r x^2 dx$$

$$= \frac{2}{r^2} \left(\frac{1}{3} x^3 \right) \Big|_0^r = \frac{2}{3} \cdot r$$

$$\text{var}(X) = E[X^2] - (E[X])^2 \quad \therefore E[X^2] = \int_0^r x \cdot \left(\frac{2x^2}{r^2}\right) dx = \frac{2}{r^2} \int_0^r x^3 dx$$

$$\begin{aligned} &= \frac{r^2}{2} - \left(\frac{2}{3}r\right)^2 = r^2 \left(\frac{1}{2} - \frac{4}{9}\right) = \frac{1}{18}r^2 - \frac{8}{18}r^2 = -\frac{7}{18}r^2 \\ &= \frac{r^2}{18} \end{aligned}$$

$$7b) S = \begin{cases} \frac{1}{X} & \text{if } X \leq t \\ 0 & \text{otherwise} \end{cases} \quad \begin{matrix} S \uparrow X \downarrow \\ S \downarrow X \uparrow \end{matrix}$$

↳ mixed random variable.

$$P(X \leq t) = \frac{\pi t^2}{\pi r^2} = \left(\frac{t}{r}\right)^2$$

$$P(S=0) = 1 - \left(\frac{t}{r}\right)^2$$

$$\text{CDF : } F_S(s) = P(S \leq s) = \begin{cases} 0 & ; \text{ if } s < 0 \\ 1 - \frac{t^2}{r^2} & ; \text{ if } 0 \leq s < \frac{1}{t} \\ ? & ; \text{ if } s > \frac{1}{t} \end{cases}$$

$S = \frac{1}{X} \leq s \Rightarrow X \geq \frac{1}{s}$

$$F_S(s) = P(\text{Situation 1}) + P(\text{Situation 2})$$

↳ outside circle ↳ inside circle

$$= P(X > t) + P\left(X \geq \frac{1}{s} \text{ and } X \leq t\right)$$

$$= P(X > t) + P\left(\frac{1}{s} \leq X \leq t\right)$$

$$P\left(\frac{1}{s} \leq X \leq t\right) = \frac{\pi t^2 - \pi \left(\frac{1}{s}\right)^2}{\pi r^2} = \frac{t^2 - \frac{1}{s^2}}{r^2}$$

$$\therefore F_S(s) = \left(1 - \left(\frac{t}{r}\right)^2\right) + \left(\frac{t^2 - \frac{1}{s^2}}{r^2}\right)$$

$$= 1 - \frac{1}{r^2 s^2}$$

$$\therefore F_S(s) = P(S \leq s) = \begin{cases} 0 & ; \text{ if } s < 0 \\ 1 - \frac{t^2}{r^2} & ; \text{ if } 0 \leq s < \frac{1}{t} \\ 1 - \frac{1}{r^2 s^2} & ; \text{ if } s > \frac{1}{t} \end{cases}$$

8) Cont. RV Y & Z .

$$\text{RV } X = pY + (1-p)Z$$

a) $f_X(x) = \frac{d}{dx} P(X \leq x)$ $f_Z(z) = \frac{d}{dz} P(Z \leq z)$
 $f_Y(y) = \frac{d}{dy} P(Y \leq y)$

By TPT: $F_X(x) = P(X \leq x) = pF_Y(x) + (1-p)F_Z(x)$
 $\therefore f_X(x) = \frac{d}{dx} [pF_Y(x) + (1-p)F_Z(x)]$
 $f_X(x) = pf_Y(x) + (1-p)f_Z(x)$

b)
$$f_X(x) = \begin{cases} p\lambda e^{\lambda x} & , \text{ if } x < 0 \\ (1-p)\lambda e^{-\lambda x} & , \text{ if } x \geq 0 \end{cases} \quad \begin{matrix} \lambda > 0 \\ 0 < p < 1 \end{matrix}$$

~~By TPT~~
 $F_Y(x) = \int_{-\infty}^x \lambda e^{\lambda x} dx = -e^{\lambda x} + C$
 $F_Z(x) = \int_{-\infty}^x \lambda e^{-\lambda x} dx = -e^{-\lambda x} + C$

$F_X(x) = \begin{cases} e^{\lambda x} & \text{for } x < 0 \end{cases}$ $X \rightarrow$ Right idea of 2x RVs

$\rightarrow$ Reattempt:

$f_Y(y) = \begin{cases} \lambda e^{\lambda y} & , \text{ if } y < 0 \\ 0 & , \text{ otherwise} \end{cases}$

$F_Y(y) = \begin{cases} e^{\lambda y} & ; \text{ if } y < 0 \\ 1 & ; \text{ if } y \geq 0 \end{cases}$

$f_Z(z) = \begin{cases} \lambda e^{-\lambda z} & , \text{ if } z \geq 0 \\ 0 & , \text{ otherwise} \end{cases}$

$F_Z(z) = \begin{cases} 0 & ; \text{ if } z < 0 \\ 1 - e^{-\lambda z} & ; \text{ if } z \geq 0 \end{cases}$

$F_Z(z) = \int_0^z \lambda e^{-\lambda x} dx = -e^{-\lambda x} \Big|_0^z = -e^{-\lambda z} - (-e^0) = 1 - e^{-\lambda z}$

By TPT $\therefore F_X(x) = pF_Y(x) + (1-p)F_Z(x)$

$$F_X(x) = \begin{cases} p e^{\lambda x} & \text{if } x < 0 \\ p e^{\lambda x} + (1-p)(1 - e^{-\lambda x}) & \text{if } x \geq 0 \end{cases}$$

• Section 3.3 Problem Set

11) $E[X] = 0$ $E[Y] = 1$
 $\text{var}(X) = 1$ $\text{var}(Y) = 4$

a) $P(X \leq 1.5) = \Phi\left(\frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{1.5-0}{\sqrt{1}}\right) = \Phi(0.75) \stackrel{x}{=} 0.77337$

$X, Y \rightarrow$ normal random variables

$\hookrightarrow \mu=0$
 $\sigma=1$

$\rightarrow P(X \leq 1.5) = \Phi(1.5) = 0.93319$

b) PDF of $(Y-1)/2$
 $\hookrightarrow$ Std Normal

$\frac{Y - \text{mean}}{\text{std dev}} \rightarrow$ recentre mean to 0
 $\rightarrow$ reduce spread to 1
Std Normal

c) $P(-1 \leq Y \leq 1) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right)$
 $= \Phi\left(\frac{1-1}{2}\right) - \Phi\left(\frac{-1-1}{2}\right)$
 $= \Phi(0) - \Phi(-1)$
 $= (0.5) - (0.15866)$
 $= 0.34134$ ✓

13) $T \rightarrow$ normal random variable

$\mu = 10^\circ\text{C}$ $\sigma = 10^\circ\text{C}$

$P(T \leq 59^\circ\text{F}) = P(T \leq 15^\circ\text{C}) = \Phi\left(\frac{15-10}{10}\right) = \Phi\left(\frac{1}{2}\right)$
 $= 0.69146$ ✓

• Section 3.4 Problem Set

15) $\{ (x, y) \mid x^2 + y^2 \leq r^2, y \geq 0 \}$ for $r > 0$.

a) Joint PDF: uniform in semi-circle

$\hookrightarrow$ Area circle = πr^2 semi = $\frac{\pi r^2}{2}$ Uniform over an area
 $\hookrightarrow$ So calc area.
 Uniform = $\frac{1}{\text{area of } S}$
 $f_{X,Y}(x, y) = \begin{cases} \frac{2}{\pi r^2} & \text{for } 0 \leq (x, y) \leq (x, y) \text{ in semi-circle} \\ 0 & \text{otherwise} \end{cases}$

b) $f_Y(y) = \int_a^b f_{X,Y}(x, y) dx$ $[a; b]$

$[a, b] \rightarrow$ Range of X

$x^2 + y^2 \leq r^2$

$x^2 + y^2 = r^2$

$x = \pm \sqrt{r^2 - y^2}$

$[-\sqrt{r^2 - y^2}; +\sqrt{r^2 - y^2}]$

$f_Y = \int_{-\sqrt{r^2 - y^2}}^{+\sqrt{r^2 - y^2}} \frac{2}{\pi r^2} dx$

$= \frac{2}{\pi r^2} x \Big|_{-\sqrt{r^2 - y^2}}^{+\sqrt{r^2 - y^2}} = \frac{2}{\pi r^2} [\sqrt{r^2 - y^2} - (-\sqrt{r^2 - y^2})]$

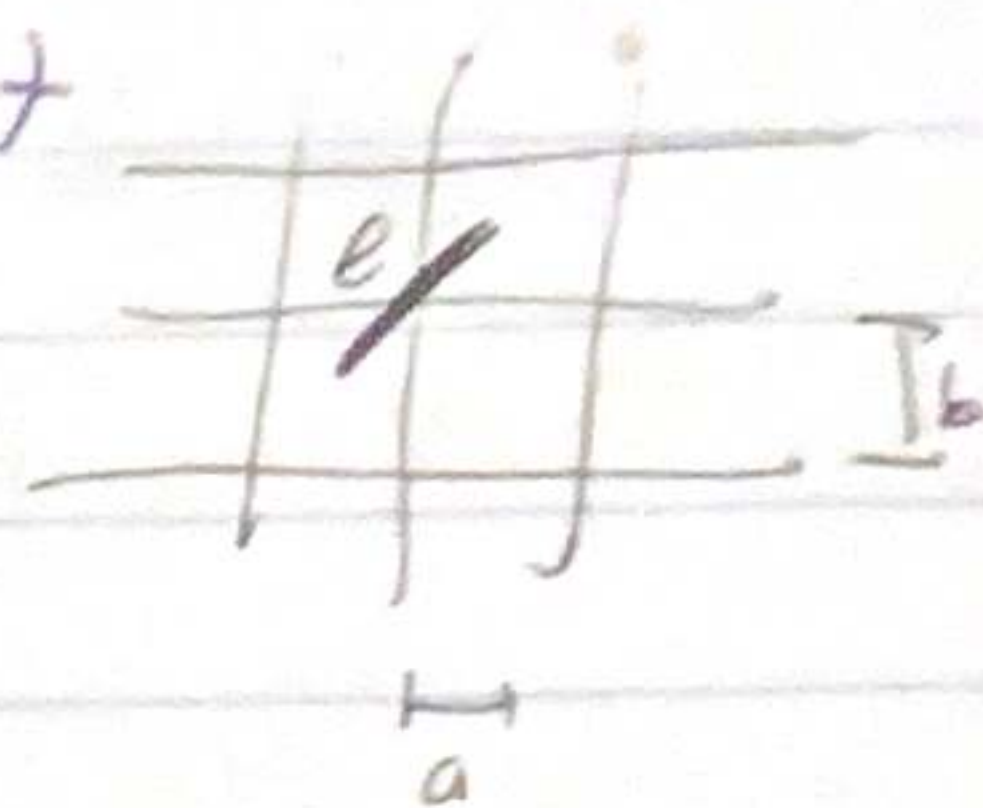
$= \frac{4\sqrt{r^2 - y^2}}{\pi r^2}$

$\therefore f_Y(y) = \begin{cases} \frac{4\sqrt{r^2 - y^2}}{\pi r^2} & ; \text{ for } 0 \leq y \leq r \\ 0 & ; \text{ otherwise} \end{cases}$

$y \geq 0$ def.
 (bottom semi)

$y \leq r \rightarrow$ want defined behavior
 (top semi)

16) Needle length = l . $\rightarrow$ Workthrough attempt
 Drop on plane with horizontal lines = a apart
 with vertical lines = b apart
 $\Rightarrow (l < a); (l < b)$



From 3.11: $f_{x,\theta}(\alpha, \theta) = \begin{cases} \frac{4}{\pi d} & , \text{ if } x \in [0, d/2] \text{ \& } \theta \in [0, \pi/2] \\ 0 & , \text{ otherwise.} \end{cases}$

$$\frac{42l}{\pi d} \Rightarrow P(A) = \frac{42l}{\pi a}$$

$$P(B) = \frac{42l}{\pi b}$$

$\hookrightarrow$ Never cross more than one line

$$E[\text{Crossed Lines}] = P(A) + P(B)$$

$$= \frac{2l}{\pi a} + \frac{2l}{\pi b}$$

$$= \frac{2lb}{\pi ab} + \frac{2la}{\pi ba}$$

$$= \frac{2l(a+b)}{\pi ab}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \underbrace{E[\text{Crossed Lines}]} - P(A \cap B)$$

$P(A \cap B) = ?$ $\rightarrow$ Use 2nd tool from 3.11: $X \leq \frac{l}{2} \sin \theta$
 $\cos \theta$ for vertical equivalent.

$$P(A \cap B) = P\left(X \leq \frac{l \sin \theta}{2}, Y \leq \frac{l \cos \theta}{2}\right)$$

...

$\hookrightarrow$ Require to determine triple integral over (X, Y, θ)

$$\hookrightarrow f_{X,Y,\theta}(x,y,\theta) = \frac{1}{\text{volume}} = \frac{1}{(\pi ab/8)} = \frac{8}{\pi ab}$$

• Section 3.5 Problem Set

18) $X \rightarrow RV$

$$PDF: f_X(x) = \begin{cases} x/4, & \text{if } 1 < x <= 3 \\ 0, & \text{otherwise} \end{cases}$$

$A \rightarrow \text{event } \{X \geq 2\}$

$$\begin{aligned} a) E[X] &= \int_1^3 x \cdot \left(\frac{x}{4}\right) dx \\ &= \frac{1}{4} \int_1^3 x^2 dx \\ &= \frac{1}{4} \left[\frac{1}{3} x^3 \right]_1^3 = \frac{1}{12} (27 - 1) \\ &= \frac{26}{12} = \frac{13}{6} \quad \checkmark \end{aligned}$$

$$\begin{aligned} P(A) &= P(2 \leq X \leq 3) \\ &= P(X \leq 3) - P(X \leq 2) \\ &= \frac{1}{4} \int_1^3 x dx - \frac{1}{4} \int_1^2 x dx \\ &= \frac{1}{4} \left[x^2 \Big|_1^3 - x^2 \Big|_1^2 \right] \\ &= \frac{1}{4} [(9 - 1) - (4 - 1)] = \frac{1}{4} [8 - 4 + 1] \\ &= \frac{5}{4} \quad \times \end{aligned}$$

$$\begin{aligned} \hookrightarrow P(A) &= \frac{1}{4} \int_2^3 x dx \\ &= \frac{1}{4} \left[\frac{1}{2} x^2 \right]_2^3 = \frac{1}{8} (9 - 4) \\ &= \frac{5}{8} \quad \checkmark \end{aligned}$$

$\hookrightarrow \int x \rightarrow \frac{1}{2} x^2 + C$

$$f_{X|A}(x) = \begin{cases} x/4, & \text{if } 2 \leq x \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

$$f_{X|A}(x) = \begin{cases} \frac{f_X(x)}{P(A)} & ; x \in A \\ 0 & ; \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{(\frac{1}{4}x)}{(\frac{5}{8})} = \frac{1}{4} \cdot x \cdot \frac{8}{5} = \frac{2x}{5} & ; \text{if } 2 \leq x \leq 3 \\ 0 & ; \text{otherwise} \end{cases}$$

$$18 \text{ con.}) E[X|A] = \int x \cdot f_{X|A}(x) dx$$

$$= \int_2^3 x \cdot \left(\frac{2x}{5}\right) dx = \frac{2}{5} \int_2^3 x^2 dx$$

$$= \frac{2}{5} \left(\frac{1}{3} \cdot x^3 \Big|_2^3 \right)$$

$$= \frac{2}{15} (27 - 8)$$

$$= \frac{38}{15} \checkmark$$

$$b.) Y = X^2$$

$$E[Y] = E[X^2] = \int_1^3 x \cdot f_X(x)^2 dx$$

$$= \int_1^3 x \cdot \left(\frac{x}{4}\right)^2 dx = \frac{1}{16} \int_1^3 x^3 dx$$

$$= \frac{1}{16} \left(x^4 \Big|_1^3 \right) = \frac{1}{16} (81 - 1)$$

$$= \frac{80}{16} = \frac{40}{8} = \frac{10}{2} = 5$$

$\xrightarrow{5/4} x \rightarrow x^2 \text{ not } f_X(x)^2$

$$= \int_1^3 x \cdot \left(\frac{x^2}{4}\right) dx$$

$$= \frac{1}{4} \int_1^3 x^3 dx = \frac{1}{4} \left(\frac{1}{4} \right) x^4 \Big|_1^3$$

$$= \frac{1}{16} \cdot (81 - 1)$$

$$= \frac{80}{16} = 5 \checkmark$$

$$\text{var}(Y) = \cancel{\text{var}(X^2)} = E[Y^2] - (E[Y])^2$$

$$= E[X^4] - (E[X^2])^2$$

$$\therefore E[X^4] = \int_1^3 x \cdot \frac{x^4}{4} dx = \frac{1}{4} \int_1^3 x^5 dx = \frac{1}{24} \left(x^6 \Big|_1^3 \right)$$

$$= \frac{1}{24} (729 - 1) = \frac{91}{3}$$

$$\therefore \text{var}(Y) = \left(\frac{91}{3} \right) - (5)^2$$

$$= \frac{16}{3} \checkmark$$

- 20) 2 appointments $\rightarrow$ durations indepe.
 $\rightarrow$ durations exponen distri.
 $\rightarrow \mu(\text{dur}) = 30 \text{ min.}$

Student 1 $\rightarrow$ on time

Student 2 $\rightarrow$ 5 min late.

~~Time between 1 arrive & 2 leave~~

$Y \rightarrow$ Appointment for Student 1 duration

$X \rightarrow$ Duration of Appointment for Student 2.

~~$f_X(x) = \begin{cases} \lambda e^{-\lambda x} & , x > 0 \\ 0 & , \text{otherwise} \end{cases}$~~

~~$f_X(x) = \begin{cases} 30 e^{-30x} & ; x > 0 \\ 0 & ; \text{otherwise} \end{cases}$~~

$\rightarrow$ Start with $E[\text{time}]$ expression

$$\Rightarrow E[\text{time}] = E[S_1] + E[S_2]$$

$$= \cancel{E} (5 + E[S_2]) \cdot P(S_1 \text{ stay} \leq 5 \text{ min}) + (E[S_1] + E[S_2]) P(S_1 \text{ stay} > 5 \text{ min})$$

$$\Rightarrow (5 + 30) P(S_1 \text{ stay} \leq 5 \text{ min}) + (2(30)) P(S_1 \text{ stay} > 5 \text{ min})$$

$$\Rightarrow (35) P(S_1 \text{ stay} \leq 5 \text{ min}) + (60) P(S_1 \text{ stay} > 5 \text{ min})$$

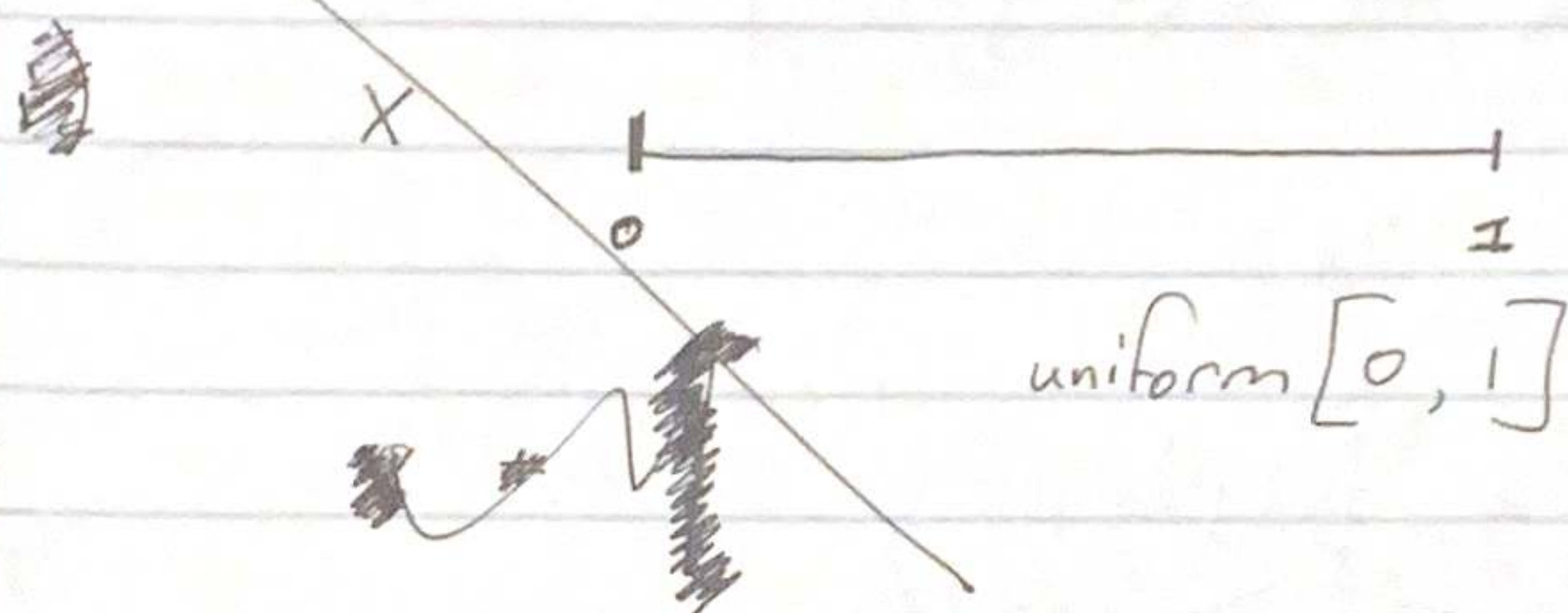
$\Rightarrow$

$$\textcircled{1} P(S_1 \text{ stay} \leq 5 \text{ min}) = \left. \begin{array}{l} F_X(x) = P(X \leq x) = \int_0^x \lambda e^{-\lambda t} dt \\ = 1 - e^{-\lambda x} \end{array} \right\}$$

$$\textcircled{2} P(S_2 \text{ stay} > 5 \text{ min}) = \left. \begin{array}{l} \textcircled{1} = 1 - e^{-(\frac{1}{30})(5)} = 0,1535 \quad \textcircled{4} \\ \textcircled{2} = e^{-(\frac{1}{30})(5)} = 0,8464 \quad \textcircled{5} \end{array} \right\}$$

$$\begin{aligned}
 20_{con}) E[\text{time}] &= (35)(P(S, \text{stay} \leq 5 \text{ min})) + (60)P(S, \text{stay} > 5 \text{ min}) \\
 &= (35)(0.1535) + (60)(0.8464) \\
 &= \cancel{56.162 \text{ min}} = \underline{60.39 \text{ min}} \quad \textcircled{1}
 \end{aligned}$$

22) Stick = unit Length
Break into 3 pieces.



23) X, Y joint PDF uniform over $\Delta (0,0); (0,1); (1,0)$

$$a) f_{X,Y}(x,y) = \frac{1}{\text{area}} = \frac{1}{(\frac{1}{2}bh)}$$



$$\text{Area} = \frac{1}{2}(1)(1) = 0.5 \therefore f_{X,Y}(x,y) = \frac{1}{0.5} = 2. \quad \hookrightarrow \boxed{x+y \leq 1}$$

$$\begin{aligned}
 b) f_Y &= \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx \\
 &= \int_0^{1-y} 2 dx \\
 &= 2x \Big|_0^{1-y} \\
 &= 2(1-y)
 \end{aligned}$$

$$0 \leq y \leq 1$$

x is at most 1 but depends on value of x .

$$\begin{aligned}
 c) f_{X|Y}(x|y) &= \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{2}{2(1-y)} \\
 &= \frac{1}{1-y}
 \end{aligned}$$

$$\begin{aligned}
 d) E[X+Y] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x+y) f_{X,Y}(x,y) dx dy \\
 &= \int_0^1 \int_0^{1-y} (x+y) \cdot \frac{1}{1-y} dx dy \\
 &= \int_0^1 \left(\frac{1}{2}x^2 + xy \right) \Big|_0^{1-y} dy \\
 &= \int_0^1 \left(\frac{1}{2}(1-y)^2 + y(1-y) \right) dy
 \end{aligned}$$

23 con.)

$$d) E[X|Y=y] = E[\text{Uniform}] = \frac{b-a}{2} \quad [0; 1-y]$$

$$= \frac{(1-y)-(0)}{2}$$

$$= \frac{1-y}{2} \quad 0 \leq y \leq 1$$

$$E[X] = \int_{-\infty}^{\infty} E[X|Y=y] \cdot f_Y(y) dy$$

$$= \int_0^{1-y} \left(\frac{1-y}{2} \right) (2(1-y)) dy$$

$$= \int_0^1 (1-y)^2 dy$$

$$= \int_0^1 \left(\frac{1-y}{2} \right) f_Y(y) dy$$

$$= \frac{1}{2} \int_0^1 f_Y(y) dy - \frac{1}{2} \int_0^1 y f_Y(y) dy$$

$$= \frac{1}{2} \int_0^1 (1-y) f_Y(y) dy$$

$$= \frac{1}{2} \underbrace{\int_0^1 f_Y(y) dy}_{\text{PPF} \rightarrow 1} - \frac{1}{2} \underbrace{\int_0^1 y f_Y(y) dy}_{\rightarrow E[Y]}$$

$$= \frac{1}{2} (1) - \frac{1}{2} E[Y]$$

$$E[X] = \frac{1 - E[Y]}{2}$$

$$e) E[X] = \frac{1 - E[Y]}{2}$$

$$\text{Symmetry} \Rightarrow E[X] = E[Y]$$

$\hookrightarrow X+Y=1$

$\hookrightarrow$ Same range

$\hookrightarrow$ Same Uni distr

$\hookrightarrow$ Same PDF

$\hookrightarrow$ Same mean/var

$$\therefore E[X] = \frac{1 - E[X]}{2} \Rightarrow 3E[X] = 1$$

$$E[X] = E[Y] = \frac{1}{3} \quad \checkmark$$